TRAINING AND SCALING LANGUAGE MODELS: FROM FIRST PRINCIPLES TO EFFICIENT SERVING

Module 9: KV-Cached Decoding

Reuse, Memory Growth, and Exact Autoregressive State

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Lecture map

1. From full-prefix recomputation to reuse
2. Cache memory and bandwidth
3. Correctness and benchmarking

Guiding question

What can autoregressive decoding reuse exactly, and how does that reuse turn compute into a memory-management problem?

What should be clear by the end?

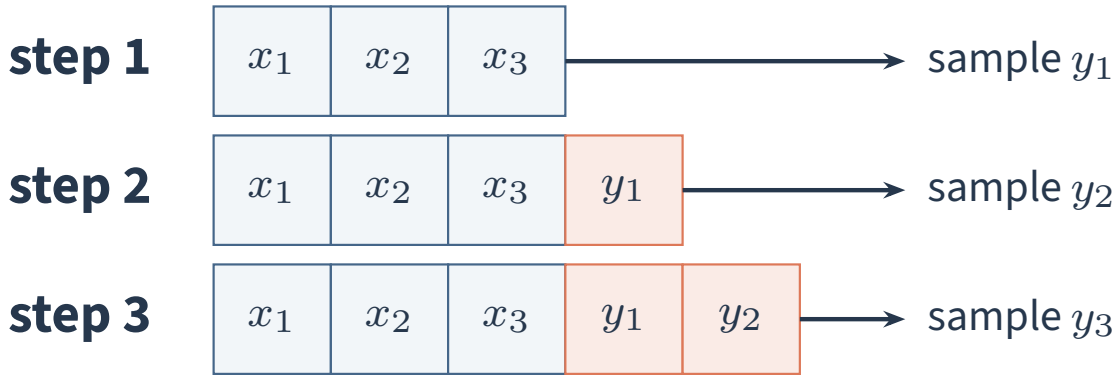
1. Explain the redundant work removed by a KV cache.
2. Compute cache shapes and memory for MHA, GQA, and MQA.
3. Reason about decode bandwidth and cache layout.
4. Test cached generation against a full-prefix reference.

PART 1

From full-prefix recomputation to reuse

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Naive decoding recomputes the unchanged prefix

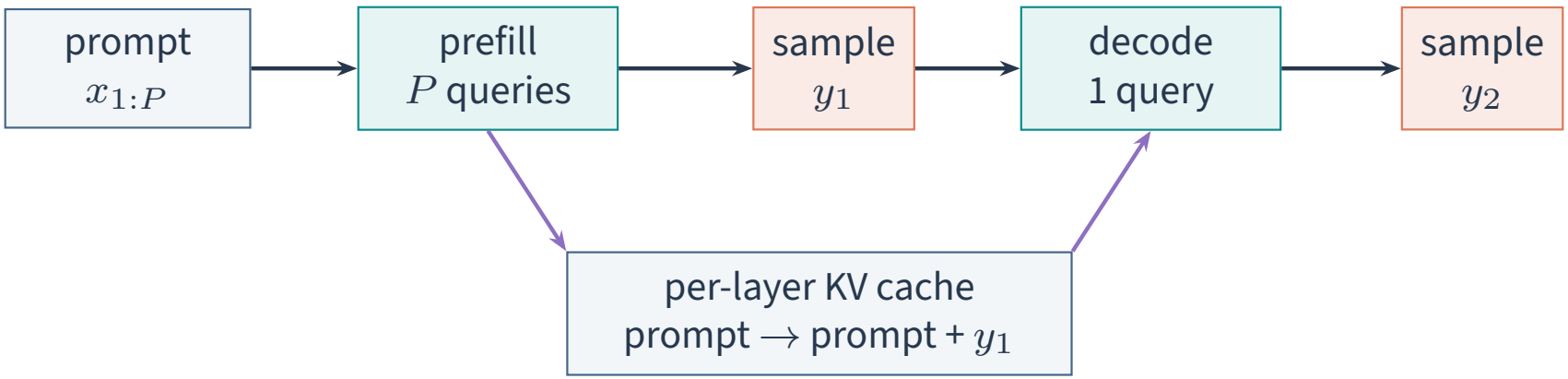


The opportunity

Causality leaves past layer states unchanged: store their keys and values instead of recomputing the full prefix at every step.

Example: three prompt tokens and two generated tokens. Sampling and model weights are fixed across the comparison.

One request: prefill produces the first token

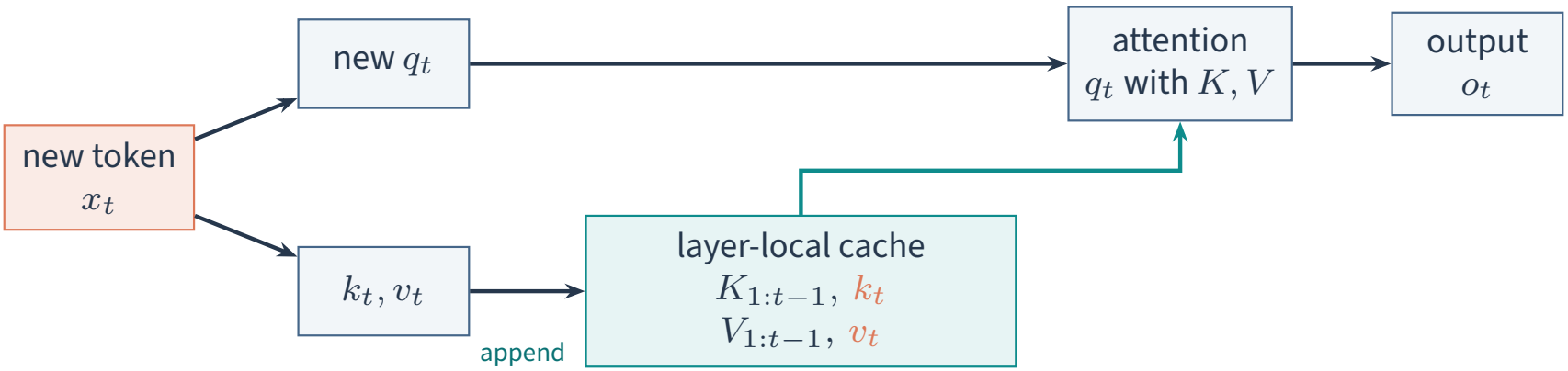


Different shapes, different bottlenecks

Prefill processes prompt tokens together. Decode consumes the previous output token and reads its growing KV history.

Prefill's final logits sample y_1; processing y_1 appends its K/V and predicts y_2.

Cached attention at one layer



$$o_t = \mathrm{softmax}\left(q_t K_{1:t}^{\top} / \sqrt{d_h}\right) V_{1:t}$$

Append this token's K/V once; its query still scans all visible cached positions.

The cache changes asymptotic projection work

For n generated tokens after a prompt:

Without cache

Reproject the growing prefix at every step. Prefix projection work accumulates roughly quadratically in generated length.

With cache

Project each token's key and value once. Attention still scans a context that grows linearly per decode step.

The cache removes redundant projection work. It does not make full-context attention constant-time.

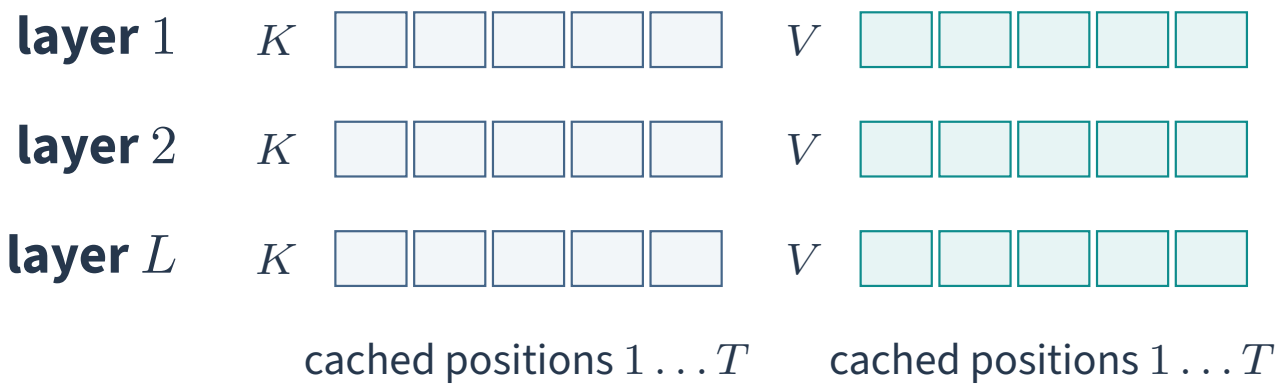
PART 2

Cache memory and bandwidth

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1. From full-prefix recomputation to reuse
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A separate cache for every layer and request



Every new token appends one K and one V vector *per layer*; each vector has $H_{kv} d_h$ elements.

Serving implication

More active requests or longer contexts mean more resident cache state. We count its bytes next.

Cache shape determines capacity

For L layers, batch B, H_{kv} key-value heads, cached length T, head width d_h, and element size s:

$$M_{KV} = 2\,L\,B\,H_{kv}\,T\,d_h\,s$$

The factor 2 stores both keys and values.

Per-token, per-request growth

$$m_{\text{token}} = 2LH_{kv}d_h s$$

This number links model architecture directly to serving concurrency.

Worked memory estimate

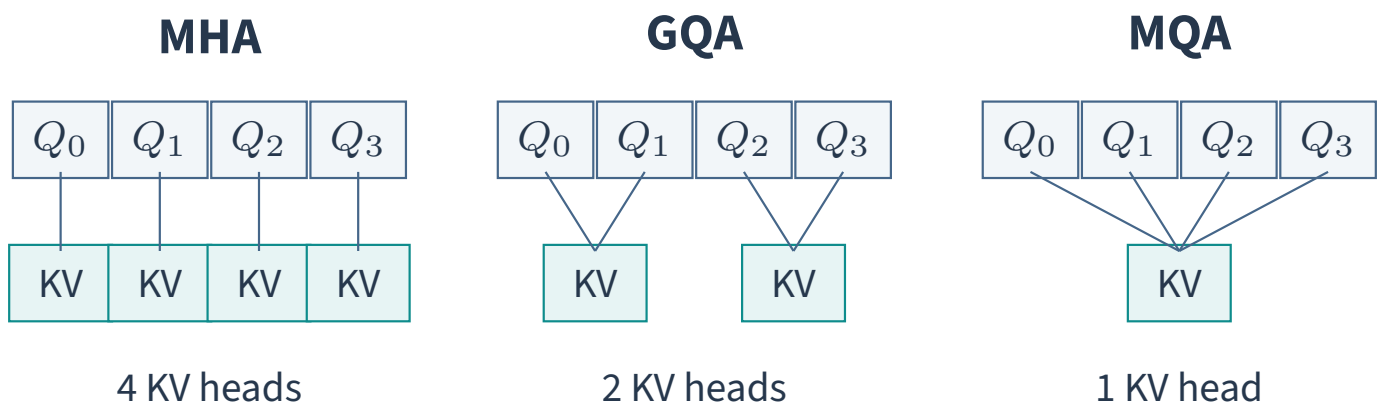
Suppose $L = 32$, $H_{kv} = 32$, $d_h = 128$, BF16, and $T = 4096$:

$$M_{KV} = 2 \cdot 32 \cdot 1 \cdot 32 \cdot 4096 \cdot 128 \cdot 2 \approx 2\,\text{GiB}$$

Serving implication

Sixteen concurrent requests at this cached length would need roughly 32 GiB for KV state alone, before weights, workspaces, and fragmentation.

MHA, GQA and MQA: who shares K/V?



Cache scales with KV heads

For this four-query-head example, relative KV memory is 1, $1/2$, and $1/4$. Query heads still have distinct queries.

Changing the sharing pattern is an architecture/training choice, not an exact reshape of an arbitrary trained MHA model.

Decode often reads far more bytes than it computes

Each new token reads the cached K, V for every active layer and request. Batch-1 decode has little reuse across queries.

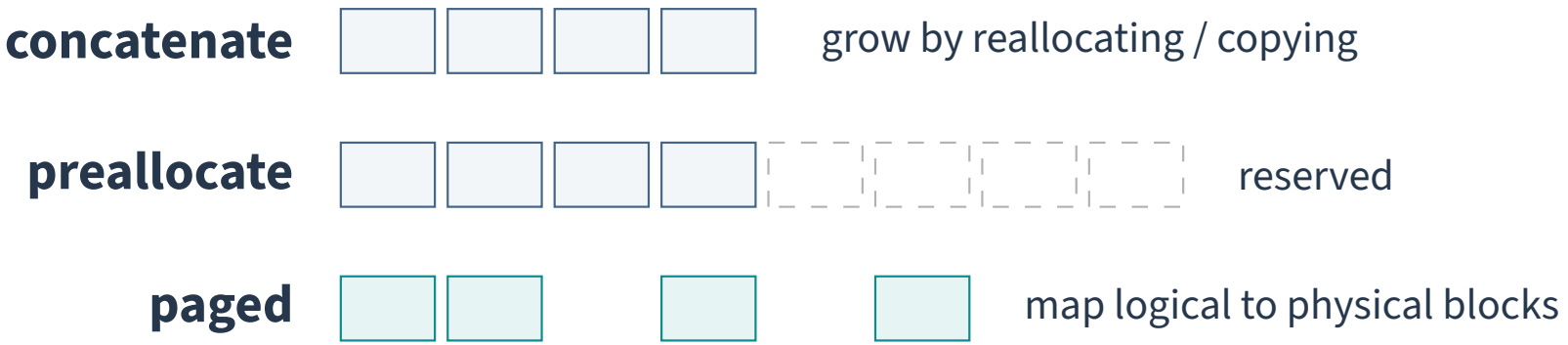
Bandwidth model

$$t_{\text{decode}} \gtrsim \frac{\text{weight bytes read} + \text{KV bytes read}}{BW_{\text{effective}}}$$

Consequence

Reducing KV heads, using lower cache precision, batching requests, and improving memory layout can matter more than reducing arithmetic.

How should the cache grow?



Concatenation may repeatedly copy; preallocation reserves capacity; paging grows in blocks and adds indirection and last-block slack.

Module 10 follows multiple requests through a shared paged allocator.

Checkpoint: capacity calculation

Think, then compare

Recompute the worked example with eight KV heads. How many 4096-token requests fit in 32 GiB if 24 GiB is available for cache after weights and workspaces?

PART 3

Correctness and benchmarking

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The cache must preserve logical positions

- The new token uses its absolute or rotary position t.
- The causal mask covers the valid cached prefix and current token.
- Padding and per-request lengths cannot expose uninitialized cache entries.
- Reordered or paged blocks must reconstruct the same logical sequence.

Common failure

Correct key-value values at the wrong positions produce plausible but divergent text.

Full-prefix decoding is the reference oracle

For fixed weights, prompt, and token sequence:

1. run a full-prefix forward pass for each step
2. run cached prefill plus single-token decode
3. compare next-token logits at every step
4. repeat across batch sizes, prompt lengths, and cache boundaries

Then test generation

Compare logits within dtype tolerance. Greedy tokens should match when the argmax is stable; sampling also needs matching probabilities and random state.

Benchmark prefill and decode separately

Phase	Primary metrics	Sweep
prefill	latency, prompt tokens/s, peak memory	prompt length, batch
decode	ms/token, output tokens/s, cache bytes	cached length, batch
end-to-end	TTFT, total latency	prompt/output distribution

Warm and synchronize

Kernel compilation, allocation, and asynchronous timing can otherwise dominate or under-report short decode steps.

Checkpoint: cache debugging

Think, then compare

Cached logits match for short prompts but diverge immediately after a page boundary. Which state and indexing checks isolate layout bugs from positional-encoding bugs?

Key takeaways

1. KV caching reuses past key and value projections exactly.
2. Cache memory grows with layers, KV heads, requests, cached tokens, head width, and precision.
3. Decode is commonly bandwidth-bound, so cache architecture and layout shape performance.
4. Full-prefix logit equivalence is the primary correctness test before benchmarking.